A Closed-Range Subcase of Dense Small Periodic Sets

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Source And Target Question

The source paper is Rodrigo Cardeccia and Santiago Muro, *Arithmetic progressions and chaos in linear dynamics*, arXiv:2003.07161. In Question 4.2 the authors ask:

> Is any hypercyclic operator with dense small periodic sets necessarily chaotic? Or, more generally, does any operator with dense small periodic sets have dense periodic points?

Question 4.1. *Does Theorem 3.11 hold on arbitrary Fréchet spaces?*

In Theorem 3.17 we showed the existence of an $\mathcal{AP}_b$-hypercyclic operator that is not chaotic. By Theorem 3.15 the operator does not have dense small periodic sets. In fact, we did not come to an operator that has dense small periodic points and does not have dense periodic points.

Question 4.2. *Is any hypercyclic operator with dense small periodic sets necessarily chaotic. Or, more generally, does any operator with dense small periodic sets have dense periodic points?*

We answered Question 1.1 for a wide class of operators and spaces. However the general question whether there exists a Furstenberg family $\mathcal{F}$ for which $\mathcal{F}$-hypercyclicity is equivalent to chaos remains open.

The following diagram shows the known implications between the concepts appearing in this article. A solid arrow means that the implication holds. A dashed arrow means that the implication holds with

The packet proves a positive Banach-space subcase under a closed-range hypothesis on the periodic equations.

Definitions

Let T be a bounded linear operator on a Banach space X. A set $Y \subset X$ is a periodic set for T if $T^k(Y) \subset Y$ for some $k \geq 1$. The operator T has dense small periodic sets if every nonempty open set $U \subset X$ contains a closed periodic set.

We use the equivalent form from Proposition 3.4 of Cardeccia–Muro: for every nonempty open set U there are $k \geq 1$ and $x \in U$ such that $T^{jk}x \in U$ for every $j \geq 1$.

Proof Intuition

The source paper turns small periodic sets into actual periodic points by placing the periodic set inside a compact convex hull and applying a fixed point theorem. The argument below replaces compactness by a closed-range metric estimate.

If an orbit of $S = T^k$ stays forever in a small ball, then the Cesaro averages of that orbit stay in the same ball and satisfy

$$(I - S)z_N = \frac{x - S^N x}{N}.$$

In a Banach space this tends to zero because the orbit is bounded. Thus each small ball contains approximate fixed points of S. When $I - S$ has closed range, approximate fixed points of S must lie close to genuine fixed points of S. Those fixed points are periodic points for T.

Lemma 1 (Closed-range estimate)**.** *Let $A : X \to X$ be a bounded linear operator on a Banach space. If $\operatorname{ran} A$ is closed, then there is a constant $C > 0$ such that*

$$\operatorname{dist}(y, \ker A) \le C\|Ay\| \qquad (y \in X).$$

Proof. The operator induced by A,

$$\widetilde{A} : X/\ker A \longrightarrow \operatorname{ran} A, \qquad \widetilde{A}(y + \ker A) = Ay,$$

is a bounded bijection between Banach spaces. Since $\operatorname{ran} A$ is closed, the open mapping theorem gives a bounded inverse. Hence

$$\|y + \ker A\|_{X/\ker A} \le C\|Ay\|.$$

This quotient norm is exactly $\operatorname{dist}(y, \ker A)$. $\square$

Theorem 1. *Let $T \in \mathcal{L}(X)$ be a bounded linear operator on a Banach space X. Assume that T has dense small periodic sets and that $\operatorname{ran}(I - T^n)$ is closed for every $n \ge 1$. Then $\operatorname{Per}(T)$ is dense in X.*

Proof. Let $U \subset X$ be nonempty and open. Choose $x_0 \in X$ and $r > 0$ such that $B(x_0, r) \subset U$. Apply dense small periodicity to the smaller ball $B(x_0, r/2)$. There are $k \ge 1$ and $x \in B(x_0, r/2)$ such that, writing $S = T^k$, one has

$$S^j x \in B(x_0, r/2) \qquad (j = 0, 1, 2, \ldots).$$

For $N \ge 1$ set

$$z_N = \frac{1}{N} \sum_{j=0}^{N-1} S^j x.$$

Because the ball $B(x_0, r/2)$ is convex, each z_N belongs to $B(x_0, r/2)$. Moreover

$$(I - S)z_N = \frac{x - S^N x}{N},$$

and therefore

$$\|(I - S)z_N\| \le \frac{\|x - x_0\| + \|S^N x - x_0\|}{N} < \frac{r}{N}.$$

By hypothesis, $I - S = I - T^k$ has closed range. The closed-range estimate gives a constant $C > 0$ such that

$$\operatorname{dist}(z_N, \ker(I - S)) \le C\|(I - S)z_N\| < \frac{Cr}{N}.$$

Choose N so large that $Cr/N < r/2$. Then some $p \in \ker(I - S)$ satisfies $\|p - z_N\| < r/2$. Since $z_N \in B(x_0, r/2)$, we get $p \in B(x_0, r) \subset U$. Finally $Sp = p$, so $T^k p = p$ and p is a periodic point of T.

Every nonempty open set contains such a point, hence $\mathrm{Per}(T)$ is dense. □

Corollary 1. *Under the same closed-range hypothesis, every hypercyclic Banach-space operator with dense small periodic sets is chaotic.*

Proof. By the theorem, the periodic points are dense. Adding hypercyclicity is exactly Devaney chaos in the linear-dynamics convention used by the source paper. □

Verification And Novelty Notes

This packet is a partial result, not a full solution of Question 4.2. It covers operators for which all periodic defect operators $I - T^n$ have closed range. The proof gives no conclusion when the relevant $I - T^k$ has nonclosed range; indeed, the mechanism then produces only approximate fixed points for T^k.

Local duplicate checks were run in the lightweight indexes for arXiv:2003.07161, "dense small periodic sets", "closed range", "approximate periodic", and "Cesaro". No existing packet for this closed-range subcase was found. The source paper itself proves the weak-star compact and weighted-shift cases, but does not state this closed-range quotient argument.

The attempted Lipschitz-free counterexample route was checked against Abbar–Coine–Petitjean, *On the dynamics of Lipschitz operators*, arXiv:2011.10800. That route is not promoted here: their examples show that linearization may create dense periodic vectors even when the base map has non-dense periodic points.

References

1. R. Cardeccia and S. Muro, *Arithmetic progressions and chaos in linear dynamics*, arXiv:2003.07161.
2. M. Abbar, Q. Coine, and C. Petitjean, *On the dynamics of Lipschitz operators*, arXiv:2011.10800.